

Arshad Mahmood

📍 Bangalore, 560037 ✉ arshadmahmood1102@gmail.com ☎ +91 7392878997

🌐 arshadmahmoodportfolio.netlify.app in arshad-mahmood-775235225 🌐 arshadvk18

About

Full-Stack Software Engineer skilled in Angular, Java Spring Boot, Python, and AWS, specializing in AI-driven, scalable enterprise applications. I blend engineering domain knowledge with software development to deliver impactful solutions in water treatment, cost-estimation platforms, and AI assistants.

Recognized as Best Performer 2024 and 1st Rank – Veolia Techathon 2024, with a strong track record in driving digital transformation. I also demonstrate leadership through mentoring interns and guiding them on best practices and real-world project execution.

Education

Institute of Engineering and Technology , Lucknow

June 2019 – June 2023

Bachelor of Technology in Chemical Engineering

- CGPA: 8.14/10

Experience

Software Engineer

Bangalore, India

Veolia Water technologies and solutions

July 2024 – Present

- Built a Cooling Tower Water Treatment Calculator using Angular 16+ with complex reactive forms and heat map visualizations.
- Developed the Chemsure application with a Tank Sizing & Cost Estimation Platform (10+ configurations, location-based cost optimization), a real-time Pricing Module, and a History Module for detailed change tracking.
- Developed serverless analytical engines using Python, FastAPI, AWS Lambda, and DynamoDB with full IaC and CI/CD automation, enabling automated water-chemistry interpretation and reducing manual engineering effort by 80.
- Led migration of multiple applications from Angular 12 → 17 using custom automation scripts (CLI/Core/Material updates, MDC migration, standalone API adoption), ensuring stable and reproducible upgrades across environments.
- Integrated with Laboratory Information Management Systems (LIMS) for real-time water chemistry analysis.
- Revamped AI assistant reporting modules with intuitive UI and scalable architecture.
- Delivered end-to-end solutions used by 500+ employees, improving efficiency across business units.
- Collaborated cross-functionally in Agile teams to implement maintainable and scalable systems.

Graduate Engineer Trainee (GET)

Bangalore, India

Veolia Water technologies and solutions

July 2023 – July 2024

- Developed a Chemical Compatibility Tool using Spring Boot + SQL, combining chemical engineering principles with programming.
- Earned appreciation for creating innovative, efficient in-house tools that reduced dependency on external vendors.
- Strengthened understanding of water treatment technologies and engineering workflows.

Intern

Bangalore, India

Suez Water Technologies and Solutions

July 2022 – Sept 2022

- Automated reporting workflows and reference lists, improving operational efficiency.
- Gained exposure to Power BI, corporate networking, and cross-team collaboration.

Projects

Cooling Tower Water Treatment Program

2025

- **Tech Stack:** Angular 16+ (Standalone Components, Reactive Forms, RxJS), PrimeNG, Bootstrap, NGX-Translate, TypeScript, SCSS, RESTful APIs, JSON, JWT Authentication
- Designed and developed a responsive calculator that optimizes chemical treatment programs based on water chemistry analysis and system parameters.
- Designed intuitive heat map visualizations to display the effectiveness of treatment.
- Integrated with LIMS to import water chemistry data, and built a real-time API-based product selection tool.
- Developed JSON-based import/export functionality for collaboration and record-keeping.

Cooling Tower Analytical Interpretation Dashboard

2025

- **Tech Stack:** Python, AWS, FastAPI
- Built a backend analytical engine to interpret cooling tower water chemistry, including cycles-of-concentration, scaling & corrosion risk, and treatment recommendations.
- Developed a serverless architecture using AWS Lambda, API Gateway, DynamoDB, and CloudFormation for fully automated IaC deployment.
- Created robust data validation, parameter normalization, and OpenAPI schemas using Python, FastAPI, and Pydantic models.
- Implemented CI/CD with CodeBuild and ECR for automatic Lambda container builds and version-controlled deployments.

AI-Powered Learning Assistant Development

2025

- **Tech Stack:** Flask, Python, Google Gemini AI API, JavaScript
- Built a personalized AI learning platform with adaptive quizzes, structured prompt engineering, and progress analytics.
- Integrated Google Gemini 1.5 Flash API with structured prompt engineering and implemented robust fallback mechanisms for seamless user experience.
- Designed RESTful Flask backend with comprehensive error handling, environment management, and CORS configuration for cross-origin requests.

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Mobile E-Commerce Application

2026

- **Tech Stack:** React Native, Supabase (PostgreSQL, Auth, Storage), TypeScript, Payment Integration, REST APIs)
- Developed a cross-platform mobile e-commerce application with product catalog, cart, secure checkout, and order management.
- Built Supabase backend with authentication, database design, and cloud storage integration.
- Integrated payment gateway and implemented admin features for product, inventory, and order management.

Tank Sizing & Cost Estimation Platform

2025

- **Tech Stack:** Angular, Ant Design, Google Sheets API
- Designed and developed a comprehensive tank sizing calculator that automates complex chemical storage calculations and cost estimations based on multiple parameters.
- Built dynamic form system supporting 10+ concurrent tank configurations with conditional rendering based

on regional requirements.

- Added location-based features to automatically find nearest warehouses and calculate shipping costs for customers.
- Developed multi-format export system (PDF, Excel, JSON) with Google Sheets integration for quote generation.
- Implemented responsive design with advanced search functionality and server-side filtering for product catalogs.

Reporting Module Revamp

2024

- **Tech Stack:** Angular, Angular Material, PrimeNG, AWS
- Collaborated across teams to design and deliver a scalable, user-friendly reporting module from scratch.
- Incorporated feedback and continuously improved functionality.

Attendance Monitoring Tool

2024

- **Tech Stack:** App Script, HTML, CSS, Google Sheets
- Built a tool for 500+ users, streamlining attendance tracking across locations.

Chemical Compatibility Tool

2023

- **Tech Stack:** HTML, CSS, JavaScript, Java Spring Boot, SQL
- Sole developer; integrated chemical engineering with programming.
- Established a strong foundation for future internal full-stack tools.

Skills & Technologies

Programming: Java, Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3, AppScript

Frameworks: Angular, Spring Boot, Flask, RxJS, PrimeNG, Angular Material, Bootstrap, React Native

Cloud & Tools: AWS (S3, Lambda, CloudFront, Cognito, CodeCommit), Power BI, Supabase, JIRA, Google Apps Script

Practices: Agile, REST API Design, AI Integration, CI/CD Pipelines, Data Visualization, Cybersecurity, Mobile Application Development, Cross Platform Development

Soft Skills: Problem-Solving, Teamwork, Communication, Leadership, Innovation

Certifications

- Generative AI for Web Developers: AI-Powered Pair Programming for Full-Stack Development
- Generative AI for Business Leaders
- Practical Cybersecurity for IT Professionals
- React Native Developer
- Cert Prep: Scrum Master
- Artificial Intelligence for Cybersecurity
- Angular Essential Training
- Python Essential Training

Awards

- Best Performer of the Year 2024
- 1st Rank in Veolia Techathon 2024

Languages

English, Hindi, Arabic (Basic)